

Histogrammar exercises

Histogrammar is a Python package that allows you to make histograms from numpy arrays, and pandas and spark dataframes.

(There is also a scala backend for Histogrammar, that is used by spark.)

You can do the exercises below after the basic tutorial.

Enjoy!

```
In [1]: %%capture
# install histogrammar (if not installed yet)

!{"sys.executable}" -m pip install histogrammar

In [2]: import histogrammar as hg

In [3]: import pandas as pd
```

Dataset

Let's first load some data!

```
In [4]: # open a pandas dataframe for use below
from histogrammar import resources

df = pd.read_csv(resources.data("test.csv.gz"), parse_dates=["date"])

In [5]: df.head(2)
```

		_id	address	age	balance	company	email	eyeColor	favoriteFruit	gender	guid	...	isActive	latitude	longitude	name	phone	...
0	565ebe61ffa6f757b5029f29	109	Monument Walk, Vicksburg, California, 6247	16	\$1,333.24	DOGSPA	tamekajoyce@dogspa.com	grey	banana	female	c06a96e4-31e2-426e-aa5a-51e3c71991aa	...	False	18.82123	-131.721235	Tameka Joyce	+1 (914) 553-2622	027
1	565ebe616e073b23430d7052	602	Harkness Avenue, Tyhee, Arizona, 5853	57	\$3,777.44	APPLICA	rosettaflynn@applica.com	brown	banana	female	6bbcaf11-a6b9-4f00-9db1-84fc11161526	...	True	-82.13461	52.720886	Rosetta Flynn	+1 (984) 525-3783	167

2 rows × 21 columns



Comparing histogram types

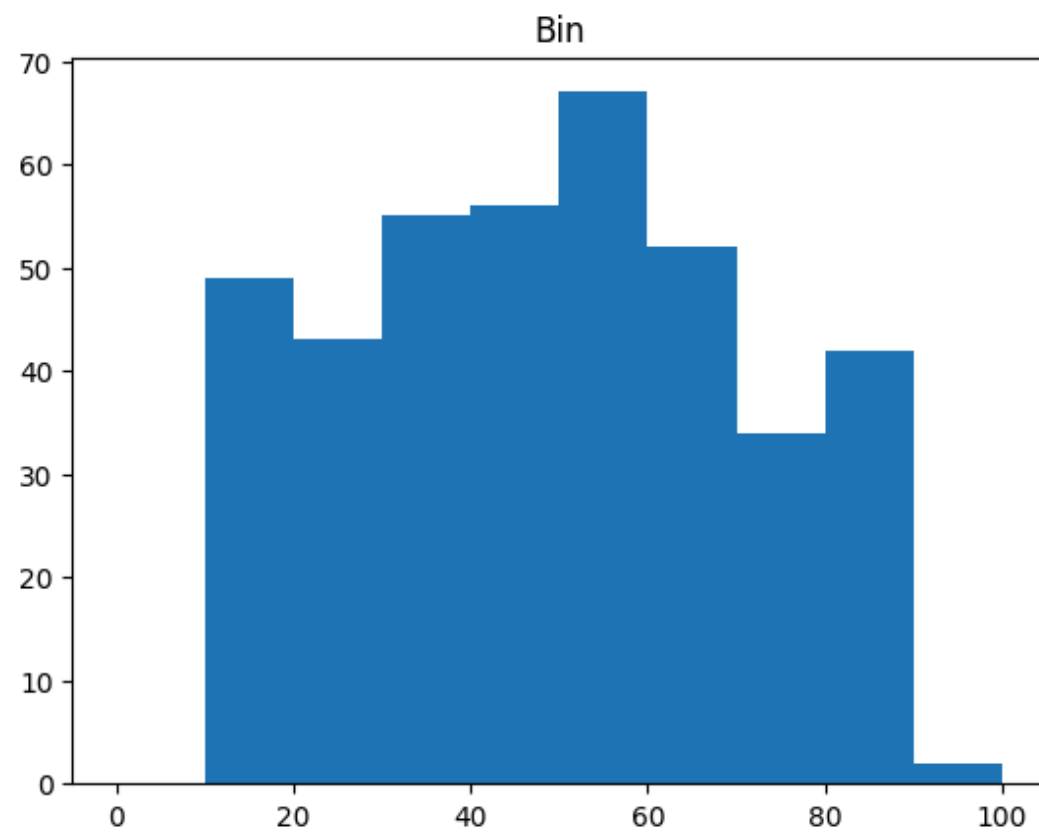
Histogrammar treats histograms as objects. You will see this has various advantages.

Let's fill a simple histogram with a numpy array.

```
In [6]: # this creates a histogram with 100 even-sized bins in the (closed) range [-5, 5]  
hist1 = hg.Bin(num=10, low=0, high=100)
```

```
In [7]: hist1.fill.numpy(df['age'].values)
```

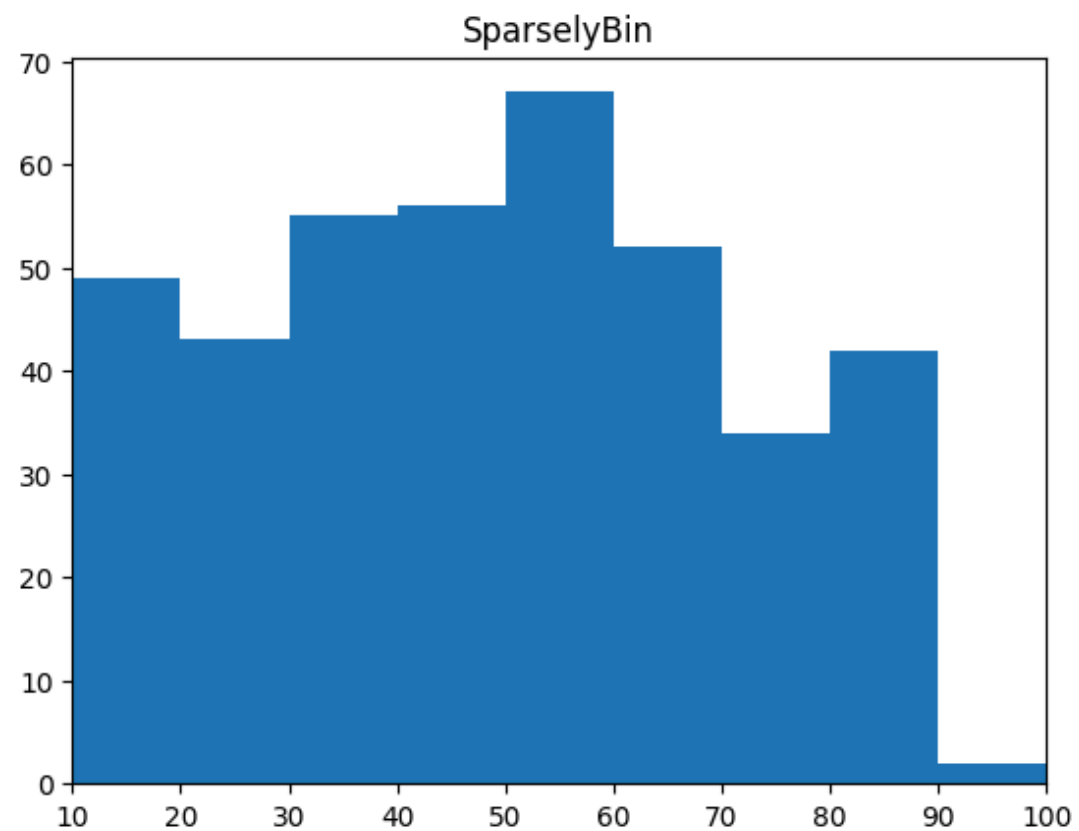
```
In [8]: hist1.plot.matplotlib();
```



```
In [9]: hist2 = hg.SparselyBin(binWidth=10, origin=0)
```

```
In [10]: hist2.fill.numpy(df['age'].values)
```

```
In [11]: hist2.plot.matplotlib();
```



Q: Have a look at the `.values` and `.bins` attributes of `hist1` and `hist2`. What types are these? (`hist1.values` is a ...?) Does that make sense?

```
In [12]: hist1
```

```
Out[12]: <Bin num=10 low=0.0 high=100.0 values=Count underflow=Count overflow=Count nanflow=Count>
```

```
In [13]: hist2
```

```
Out[13]: <SparselyBin binWidth=10.0 bins=Count nanflow=Count>
```

Q: In each bin, what type of object is keeping track of the bin count?

Try filling `hist1` with small values (negative) or very large (> 100) or with NaNs. Find out if and how `hist1` keeps track of these?

Now fill `hist2` with small values (negative) or very large (> 100) or with NaNs. How does `hist2` keeps track of these?

Categorical variables

For categorical variables use the `Categorize` histogram

- `Categorize` histograms: accepting categorical variables such as strings and booleans.

```
In [14]: histx = hg.Categorize('eyeColor')
```

```
In [15]: histx.fill.numpy(df)
```

Q: A categorize histogram, what is it fundamentally, a dictionary or a list?

Q: What else can it keep track of, e.g. numbers, booleans, nans? Give it a try, fill it with more entries!

Fill a histograms with a boolean array (isActive), directly from the dataframe

Q: what type of histogram do you get?

```
In [16]: hists = df.hg_make_histograms(features=['isActive'])
```

0%|

| 0/1 [00:00<?, ?it/s]

100%|

| 1/1 [00:00<00:00, 1461.94it/s]

```
In [ ]:
```

Multi-dimensional histograms

Let's make a 3-dimensional histogram, with axes: x=favoriteFruit, y=gender, z=isActive. (In Histogrammar, a multi-dimensional histogram is composed as recursive histograms, starting with the last one.) Then fill it with the dataframe.

```
In [17]: # hist1 = hg.Categorize(quantity='isActive')
# hist2 = hg.Categorize(quantity='gender', value=hist1)
# hist3 = hg.Categorize(quantity='favoriteFruit')
```

Q: How many data points end up in the bin: banana, male, True ?

Q: Store this histogram as a json file. What is the size of the json file?

Q: Read back the histogram and then plot it.

Q: Make a histogram of the feature 'fruit', which measures the average value of 'latitude' per bin of fruit.

```
In [18]: hist1 = hg.Average(quantity='latitude')
```

Q: what is the mean value of latitude for the bin 'strawberry'?